UVI Ocean Glider Lab Piloting Training
Day 2

Authors: Joshua Soll and Travis Hamlin
Adapted from: Slocum Training, Joe Gradone (Rutgers U)

Questions from Last Class?



Roadmap

Software Hierarchy

Glider Programming Files (.mi, .ma)

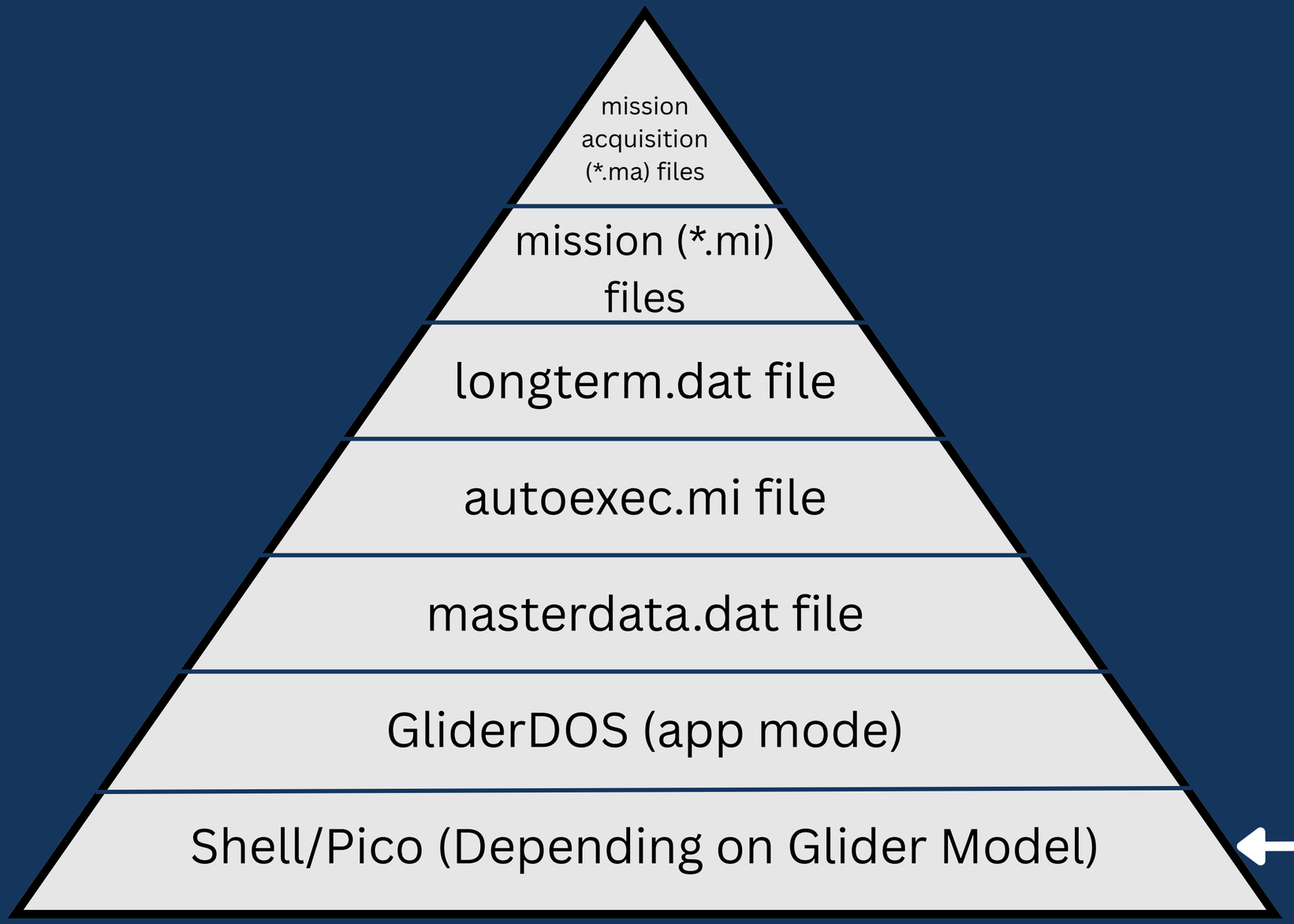
Sensor Prefixes

Glider Commands

Data Files

Glider Rules

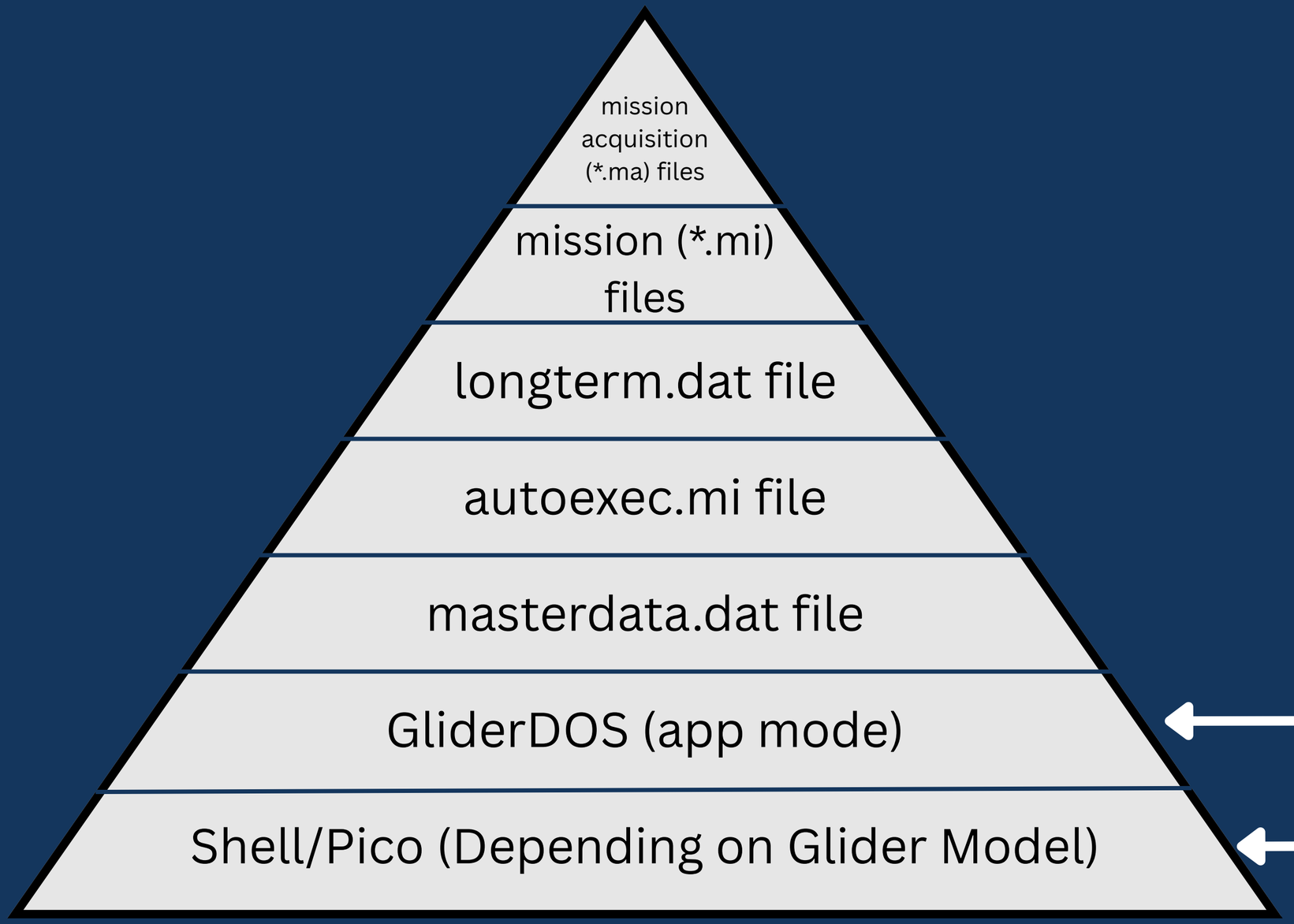
Software Hierarchy



Glider Software Hierarchy (Bottom to Top)

STM 32/Persistor operating system (basic computer)

Software Hierarchy

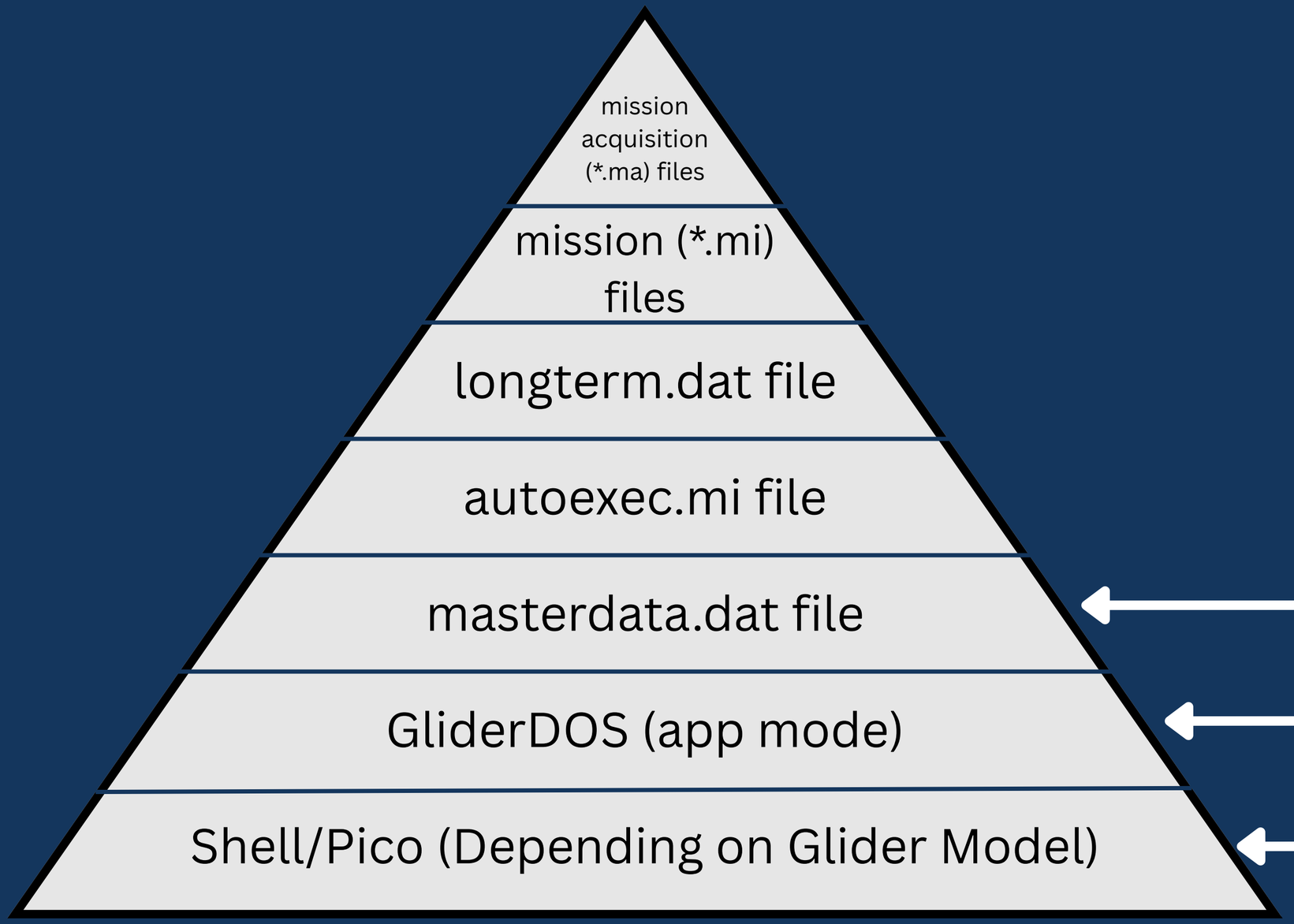


Glider Software Hierarchy (Bottom to Top)

Glider operating system

STM 32/Persistor operating system (basic computer)

Software Hierarchy



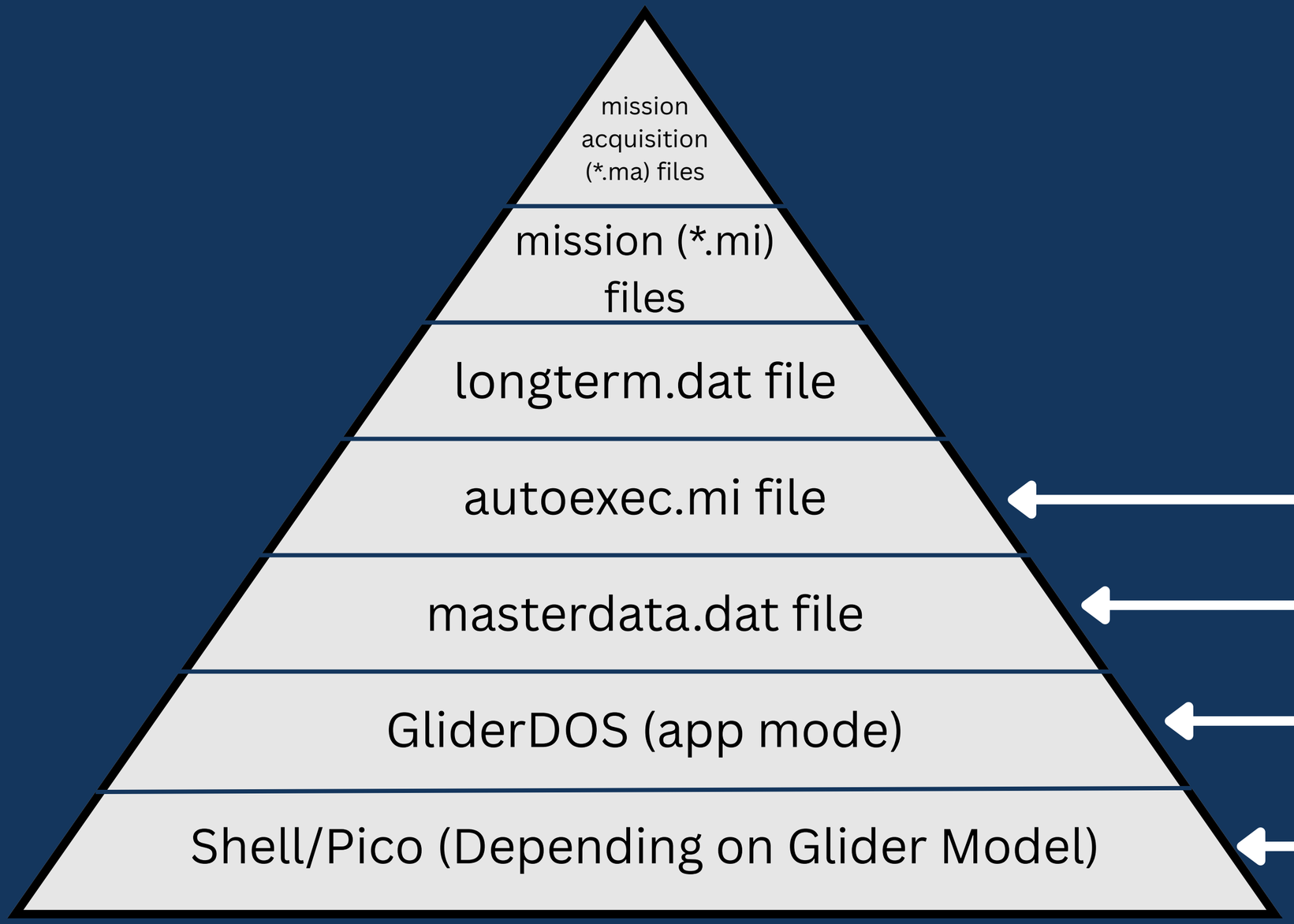
Glider Software Hierarchy (Bottom to Top)

Defines all default glider variables and argument value options

Glider operating system

STM 32/Persistor operating system (basic computer)

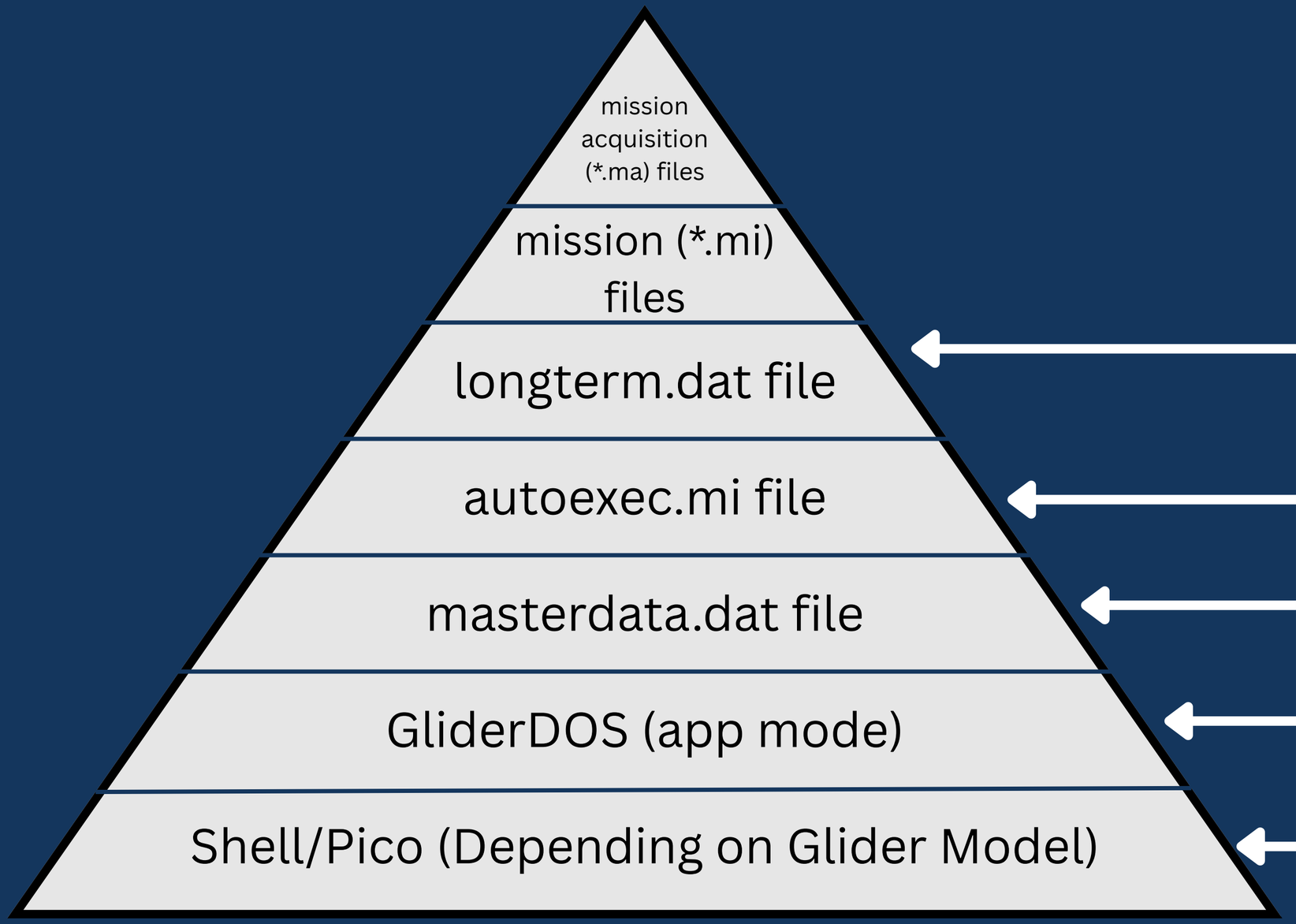
Software Hierarchy



Glider Software Hierarchy (Bottom to Top)

autoexec.mi file	**Defines glider specific variables**
masterdata.dat file	**Defines all default glider variables and argument value options**
GliderDOS (app mode)	**Glider operating system**
Shell/Pico (Depending on Glider Model)	**STM 32/Persistor operating system (basic computer)**

Software Hierarchy



Maintains values set for specific variables on power cycle

Defines glider specific variables

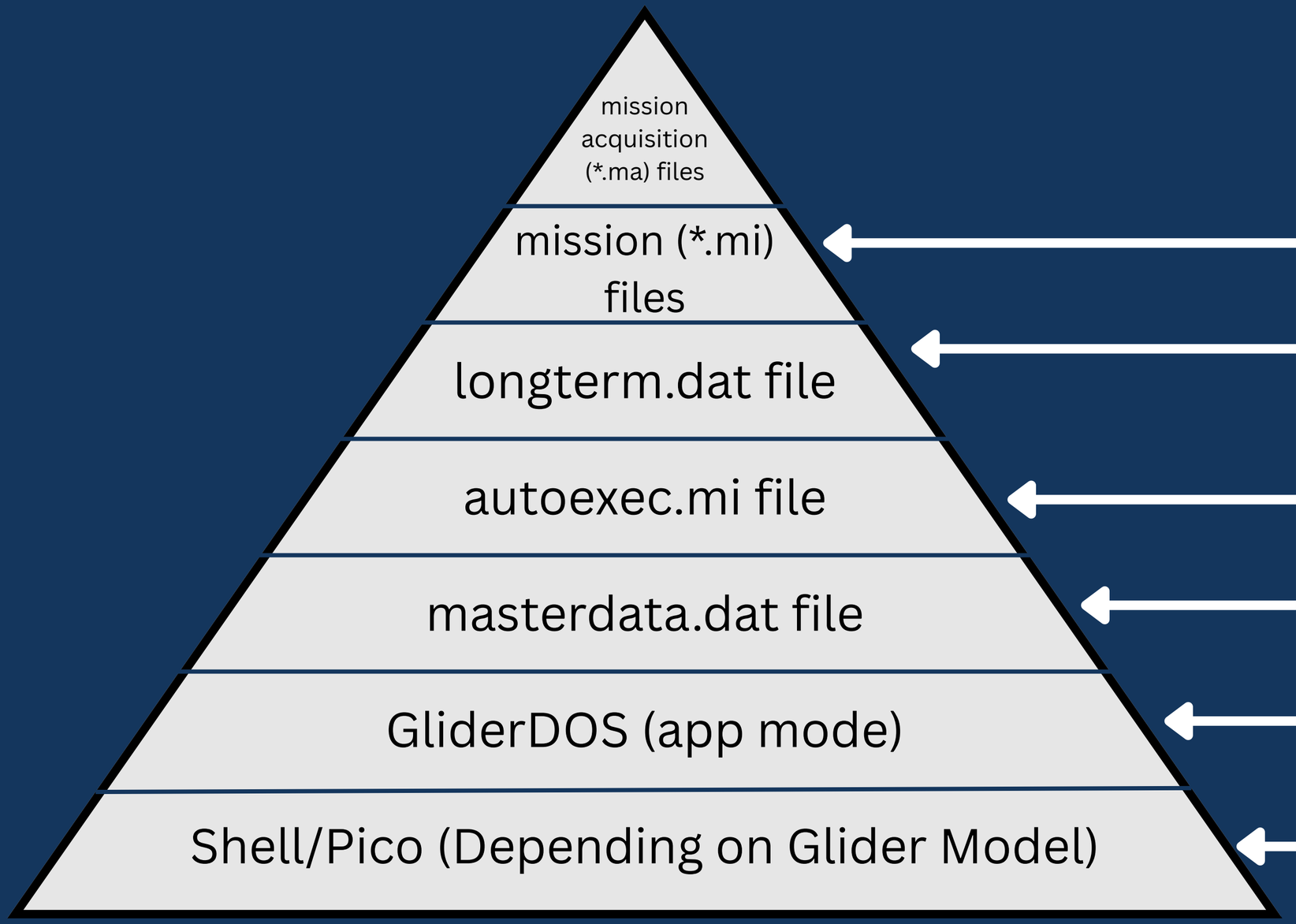
Defines all default glider variables and argument value options

Glider operating system

STM 32/Persistor operating system (basic computer)

Glider Software Hierarchy (Bottom to Top)

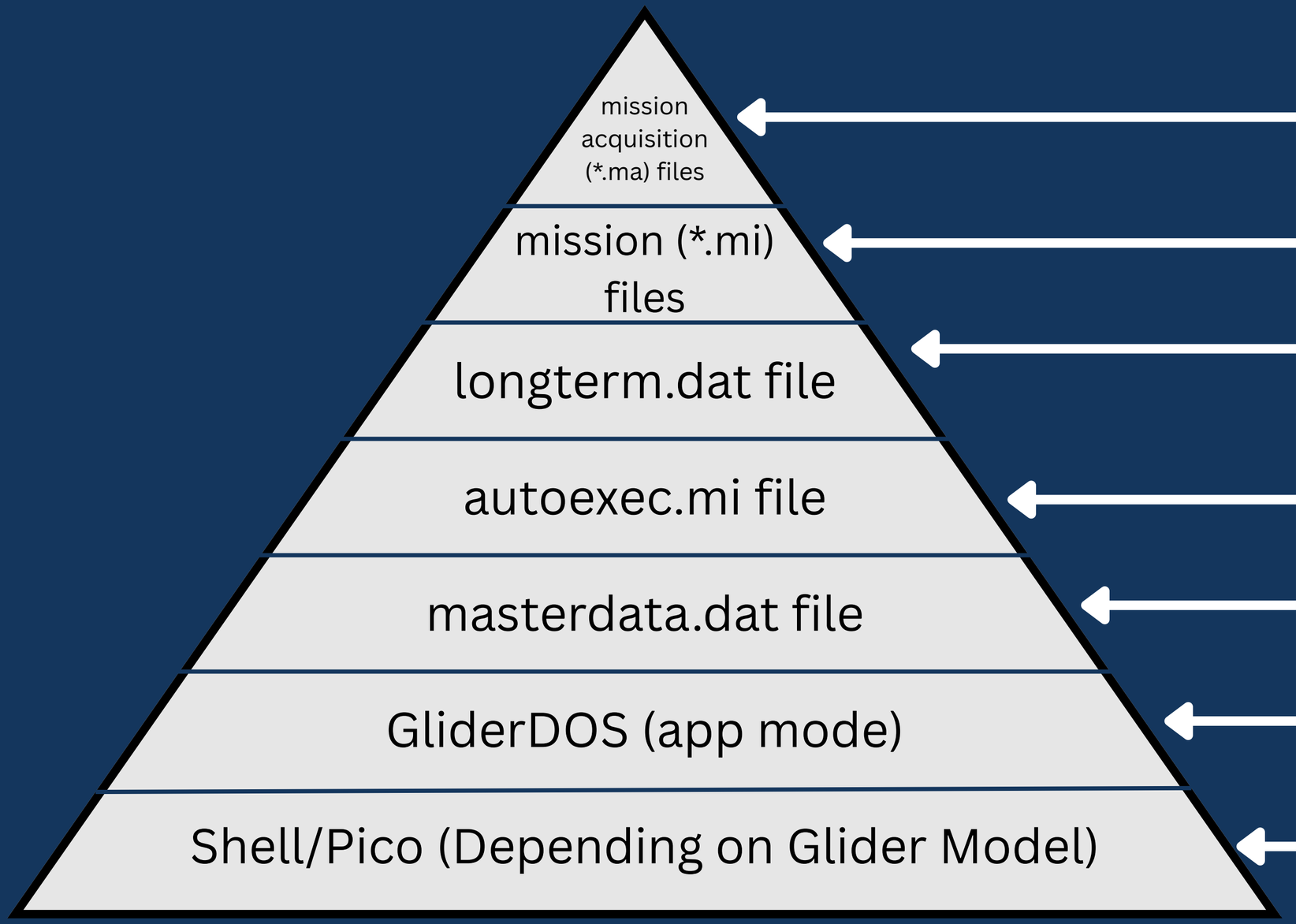
Software Hierarchy



Glider Software Hierarchy (Bottom to Top)

Defines mission variable or behavior
Maintains values set for specific variables on power cycle
Defines glider specific variables
Defines all default glider variables and argument value options
Glider operating system
STM 32/Persistor operating system (basic computer)

Software Hierarchy



Glider Software Hierarchy (Bottom to Top)

Defines mission behavior

Defines mission variable or behavior

Maintains values set for specific variables on power cycle

Defines glider specific variables

Defines all default glider variables and argument value options

Glider operating system

STM 32/Persistor operating system (basic computer)

Mission (*.mi) Files

- Defines glider behaviors while deployed in the field or simulating

Mission (*.mi) Files

- Defines glider behaviors while deployed in the field or simulating
- Mostly “MISSIONNAME.mi” format

Mission (*.mi) Files

- Defines glider behaviors while deployed in the field or simulating
- Mostly “MISSIONNAME.mi” format
- Can define arguments within this file (i.e., when to abort, when to surface), or can reference other files that define these behaviors

Mission (*.mi) Files

- Defines glider behaviors while deployed in the field or simulating
- Mostly “MISSIONNAME.mi” format
- Can define arguments within this file (i.e., when to abort, when to surface), or can reference other files that define these behaviors
- Referencing other files makes it easier to make small changes while in mission

Mission (*.mi) Files

- Defines glider behaviors while deployed in the field or simulating
- Mostly “MISSIONNAME.mi” format
- Can define arguments within this file (i.e., when to abort, when to surface), or can reference other files that define these behaviors
- Referencing other files makes it easier to make small changes while in mission
- Usually a mixture of in-line arguments and referrals to other specific files

Mission (*.mi) Files

- Defines glider behaviors while deployed in the field or simulating
- Mostly “MISSIONNAME.mi” format
- Can define arguments within this file (i.e., when to abort, when to surface), or can reference other files that define these behaviors
- Referencing other files makes it easier to make small changes while in mission
- Usually a mixture of in-line arguments and referrals to other specific files
- To edit a mission file you must stop the glider from its programmed task

Example: If you change the yo file, make sure you adjust the surfac files as needed

Example:

```
# ABORT BEHAVIOR
behavior: abend
   # OVERDEPTH: glider finds itself in > 1015 m of water
   b_arg: overdepth(m)                 1015
   b_arg: overdepth_sample_time(sec)      30

   # OVERTIME: disabled
   b_arg: overtime(sec)                -1.0

# GOTO_LIST : Waypoint list
behavior: goto_list
   b_arg: args_from_file(enum)    10  # mafiles/goto_l10.ma
   b_arg: start_when(enum)        0  # 0-immediately



# YO : Flight params
behavior: yo
   b_arg: args_from_file(enum)    20  # mafiles/yo20.ma
   b_arg: start_when(enum)        2  # 2-depth idle
   b_arg: end_action(enum)        2  # 2 resume
```

Mission (*.ma) Acquisition Files

- Defines specific glider behaviors (one type of behavior in a file)

Mission (*.ma) Acquisition Files

- Defines specific glider behaviors (one type of behavior in a file)
- Examples: yo.ma, surfac.ma, goto.ma

Mission (*.ma) Acquisition Files

- Defines specific glider behaviors (one type of behavior in a file)
- Examples: yo.ma, surfac.ma, goto.ma
- These are the files that can be referenced in a mission (*.mi) file.

Mission (*.ma) Acquisition Files

- Defines specific glider behaviors (one type of behavior in a file)
- Examples: yo.ma, surfac.ma, goto.ma
- These are the files that can be referenced in a mission (*.mi) file.
- MA files instead can be uploaded to the glider without stopping the glider

Example:

```
behavior_name=yo

<start:b_arg>
 # HALF-YO'S (2 = 1 yo)
 b_arg:  num_half_cycles_to_do(nodim)    2

   ### DIVE_TO ARGUMENTS ###
 b_arg:  d_target_depth(m)        75
 b_arg:  d_target_altitude(m)     60
 # Ballast Pump (absolute)
  b_arg:  d_use_bpump(enum)   2
  b_arg:  d_bpump_value(X)   -300.0
 # Dive Angle
  b_arg:  d_use_pitch(enum)    3  # 3 = AP, 1 = battpos
  b_arg:  d_pitch_value(X)     -0.454
 # Thruster
  # b_arg: d_use_thruster(enum)  0  # 0=off 2=% voltage 3=depthrate 4=1-9Watt
  # b_arg: d_thruster_value(X)    0.0
  # b_arg: d_depth_rate_method(enum) 3  # 3=average
    # STUCK SCENARIO
 b_arg:  d_stop_when_stalled_for(sec)    180
 b_arg:  d_stop_when_hover_for(sec)     180

  #b_arg: c_thruster_value(X)   0.0
    # STUCK SCENARIO
 b_arg:  c_stop_when_stalled_for(sec)    180
 b_arg:  c_stop_when_hover_for(sec)     180
<end:b_arg>
```

Sensor Prefixes

- Condensed way to describe command/action of a sensor value (i.e., read current measured value reading or command sensor to change its value)

Prefix	Significance of the Prefix and Example
m_	Measured; i.e., m_battery; <get m_battery> = tell me the glider’s current battery voltage

Sensor Prefixes

- Condensed way to describe command/action of a sensor value (i.e., read current measured value reading or command sensor to change its value)

Prefix	Significance of the Prefix and Example
m_	Measured; i.e., m_battery; <get m_battery> = tell me the glider's current battery voltage
c_	Commanded; i.e., c_battpos; <put c_battpos 0.4> = move the pitch battery forward to 0.4 inches

Sensor Prefixes

- Condensed way to describe command/action of a sensor value (i.e., read current measured value reading or command sensor to change its value)

Prefix	Significance of the Prefix and Example
m_	Measured; i.e., m_battery; <get m_battery> = tell me the glider's current battery voltage
c_	Commanded; i.e., c_battpos; <put c_battpos 0.4> = move the pitch battery forward to 0.4 inches
u_	User-defined before run-time; i.e., u_max_time_in_gliderdos; <put u_max_time_in_gliderdos 3600> = allow glider to stay in GliderDOS for an hour before attempting to resume mission

Sensor Prefixes

- Condensed way to describe command/action of a sensor value (i.e., read current measured value reading or command sensor to change its value)

Prefix	Significance of the Prefix and Example
m_	Measured; i.e., m_battery; <get m_battery> = tell me the glider's current battery voltage
c_	Commanded; i.e., c_battpos; <put c_battpos 0.4> = move the pitch battery forward to 0.4 inches
u_	User-defined before run-time; i.e., u_max_time_in_gliderdos; <put u_max_time_in_gliderdos 3600> = allow glider to stay in GliderDOS for an hour before attempting to resume mission
f_	Value set in factory; i.e., f_coulomb_battery_capacity

Sensor Prefixes

- Condensed way to describe command/action of a sensor value (i.e., read current measured value reading or command sensor to change its value)

Prefix	Significance of the Prefix and Example
m_	Measured; i.e., m_battery; <get m_battery> = tell me the glider's current battery voltage
c_	Commanded; i.e., c_battpos; <put c_battpos 0.4> = move the pitch battery forward to 0.4 inches
u_	User-defined before run-time; i.e., u_max_time_in_gliderdos; <put u_max_time_in_gliderdos 3600> = allow glider to stay in GliderDOS for an hour before attempting to resume mission
f_	Value set in factory; i.e., f_coulomb_battery_capacity
x_	Computed at run time; i.e., x_last_wpt_lon

Sensor Prefixes

- Condensed way to describe command/action of a sensor value (i.e., read current measured value reading or command sensor to change its value)

Prefix	Significance of the Prefix and Example
m_	Measured; i.e., m_battery; <get m_battery> = tell me the glider's current battery voltage
c_	Commanded; i.e., c_battpos; <put c_battpos 0.4> = move the pitch battery forward to 0.4 inches
u_	User-defined before run-time; i.e., u_max_time_in_gliderdos; <put u_max_time_in_gliderdos 3600> = allow glider to stay in GliderDOS for an hour before attempting to resume mission
f_	Value set in factory; i.e., f_coulomb_battery_capacity
x_	Computed at run time; i.e., x_last_wpt_lon
s_	Variables used in simulation; i.e., s_battery (simulated voltage use)

Sensor Prefixes

- Condensed way to describe command/action of a sensor value (i.e., read current measured value reading or command sensor to change its value)

Prefix	Significance of the Prefix and Example
m_	Measured; i.e., m_battery; <get m_battery> = tell me the glider's current battery voltage
c_	Commanded; i.e., c_battpos; <put c_battpos 0.4> = move the pitch battery forward to 0.4 inches
u_	User-defined before run-time; i.e., u_max_time_in_gliderdos; <put u_max_time_in_gliderdos 3600> = allow glider to stay in GliderDOS for an hour before attempting to resume mission
f_	Value set in factory; i.e., f_coulomb_battery_capacity
x_	Computed at run time; i.e., x_last_wpt_lon
s_	Variables used in simulation; i.e., s_battery (simulated voltage use)
sci_	Science sensor; i.e., sci_water_temp

Sensor Prefixes

```
Glider uvi_01 at surface.
Because:nothing commanded [behavior surface_2 start_when = 1.0]
MissionName:abc_adcp.mi MissionNum:uvi_01-2026-047-0-11 (0082.0011)
Vehicle Name: uvi_01
Curr Time: Wed Feb 18 14:20:55 2026 MT:   69837
DR  Location:  1320.060 N -6901.095 E measured    707.334 secs ago
GPS TooFar:   69696969.000 N 69696969.000 E measured     1e+308 secs ago
GPS Invalid :  1321.688 N -6901.115 E measured    765.457 secs ago
GPS Location:  1320.060 N -6901.095 E measured    709.995 secs ago
   sensor:c_wpt_lat(lat)=1310.029                  38655.2 secs ago
   sensor:c_wpt_lon(lon)=-6900                     38655.2 secs ago
   sensor:m_battery(volts)=14.9452554423191         40.327 secs ago
   sensor:m_coulomb_amphr(amp-hrs)=146.588574000008        3.185 secs ago
   sensor:m_coulomb_amphr_total(amp-hrs)=146.502324000008        3.189 secs ago
   sensor:m_depth(m)=0.899689392947667               3.051 secs ago
   sensor:m_digifin_leakdetect_reading(nodim)=1022         3.55 secs ago
   sensor:m_gps_mag_var(rad)=0.191986217719376      710.041 secs ago
   sensor:m_iridium_attempt_num(nodim)=0           590.573 secs ago
   sensor:m_iridium_call_num(nodim)=643              666.8 secs ago
   sensor:m_iridium_dialed_num(nodim)=935          674.692 secs ago
   sensor:m_leakdetect_voltage(volts)=2.47448107448107       40.223 secs ago
   sensor:m_leakdetect_voltage_forward(volts)=2.48791208791209       40.187 secs ago
   sensor:m_leakdetect_voltage_science(volts)=2.49169719169719       40.152 secs ago
   sensor:m_tot_num_inflections(nodim)=1198       783.524 secs ago
   sensor:m_vacuum(inHg)=9.36752989010989           40.331 secs ago
```

Common Glider Commands in GliderDOS

Name	Use	Name	Use
help	List all available commands	whoru	get glider’s name
lab_mode	enter/exit lab mode	where	get glider’s last known position
use	see what is in use/install/uninstall devices	zero_ocean_pressure	zero glider’s pressure sensor
wiggle	move all parts on glider	dir	list file menu/navigate
report ++	report sensor value everytime it changes	run	run mission
get	report sensor value once	loadmission	load variables/run specific sensor
put	change sensor value	exit	exit to shell/pico
! (bang)	use in front of any command in mission	exit reset	exit and reset the glider
consci	go to science computer		

File Transfer Commands: To Glider

Command	Example	Use
zs/!zs	<zs path/filename>	Glider sends file from specified path from flight or science computer
szs/!szs	<szs path/filename>	Glider sends file from specified path on science processor (command made from flight processor
send	<send logfilenames> <send -num=3 *.tbd *.sbd>	Glider send specified log file names (don’t need to specify path)/ Glider send 3 most recent .tbd and .sbd files
s	<s logfilenames> <s -num=1 *.tbd *.sbd>	same as “send” but used when glider is in a mission

File Transfer Commands: From Glider

Command	Example	Use
zr	<zr>	Glider receives file, opens up computer file system to select when not using SFMC/dockserver
dockzr/!dockzr	<dockzr filename>	Glider receives file from “to-glider” folder in SFMC
szr	<szr>	Science processor receives file, only used when not using SFMC/dockserver
dockszr	<dockszr filename>	Science processor receives file from “to-science” folder in SFMC

Data Files

	Flight	Science	New Compressed Files
All Data	*.DBD	*.EBD	*.DCD/*.ECD
Small Data Files (User Defined, some fraction of total data)	*.SBD	*.TBD	*.SCD/*.TCD
Medium Data Files (User Defined, some fraction of total data)	*.MBD	*.NBD	*.MCD/*.NCD
Log Files	*.MLG	*.NLG	*.MCG/*.NCG

Rules for Gliders in the Field

- Always check the vacuum before putting the glider in ANY water
- Certain things used in the lab should NOT be used in the field:
  - PICO/shell
  - use (use with extreme caution!)
  - lab_mode
  - loadmission lowpow.mi
- Be aware of glider’s surroundings in the field (land, seamounts, general bathymetry)
- Avoid piloting the glider perpendicular into shallowing slopes
- Type commands in slowly to avoid typos, and don’t type too many commands at once

Questions?

Thank you!

joshua.soll@students.uvi.edu
travis.hamlin@uvi.edu